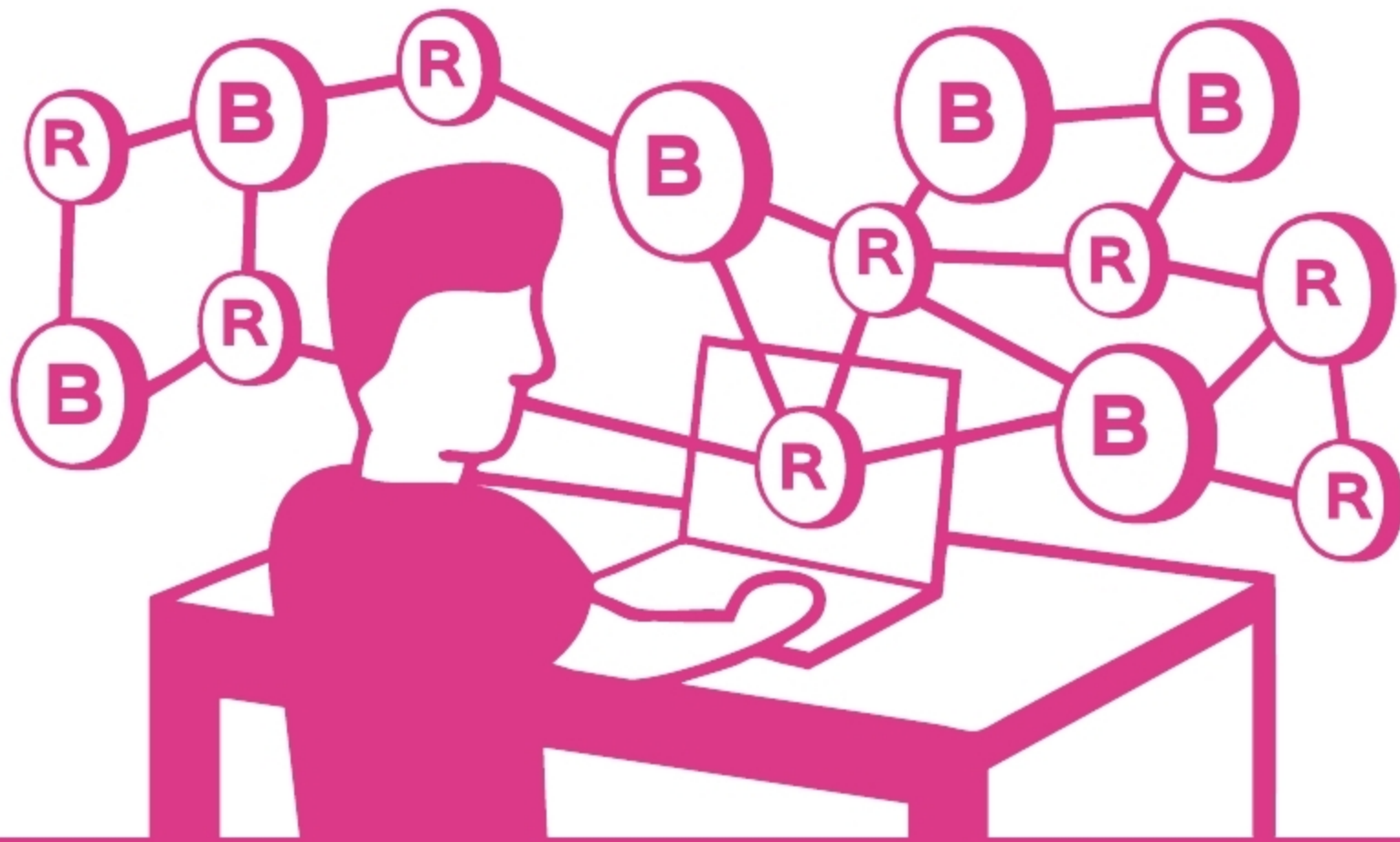


Building a Blockchain in the R Language



It is rather difficult to understand how blockchain technology works. Blockchain in R is an effective way to understand programming and the crypto principles behind this revolutionising technology. R programming involves several objects that enable one to understand blockchain in a simpler way. It also contains several packages related to blockchain that make the entire implementation more effective.

Bitcoin, an innovative payment network, has brought fame to blockchain technology. Blockchain is considered a distributed software network that comprises a shared and trusted ledger of transactions. It allows users to trust the output of the system without trusting any actor within it by providing a universal state layer. It maintains a continuously growing list of transactions and chained blocks that are cryptographically secure from any kind of tampering. The structure of this ledger is a linked list or chain of blocks. Each block in this

architecture contains a certain number of transactions that are validated by the network in a given time span. Each block also has a cryptographic hash of the prior block in the blockchain. Practically, it is very difficult to understand the blockchain at first glance.

The R programming language makes it easier to understand blockchains for even novices. The list object of the R programming language creates a block very easily. We can add different kinds of elements to this object such as an identifier, timestamp, data, parent record, and

so on. With the help of the package in R, we are able to generate hash codes that are essential to each block in the blockchain. R programming provides the concept of the function that is useful for generating validation and the 'proof of work' kind of functionalities of blockchain.

Building a blockchain in R

Creating a block in the blockchain:

A block is the starting point for the blockchain implementation. The object 'list' of R plays a role in the creation of a block.

